ADVANCE 5G AND 6G COMMUNICATION SYSTEM

EXPERIMENT 5

OBJECTIVE 1:

PROGRAM CODE :

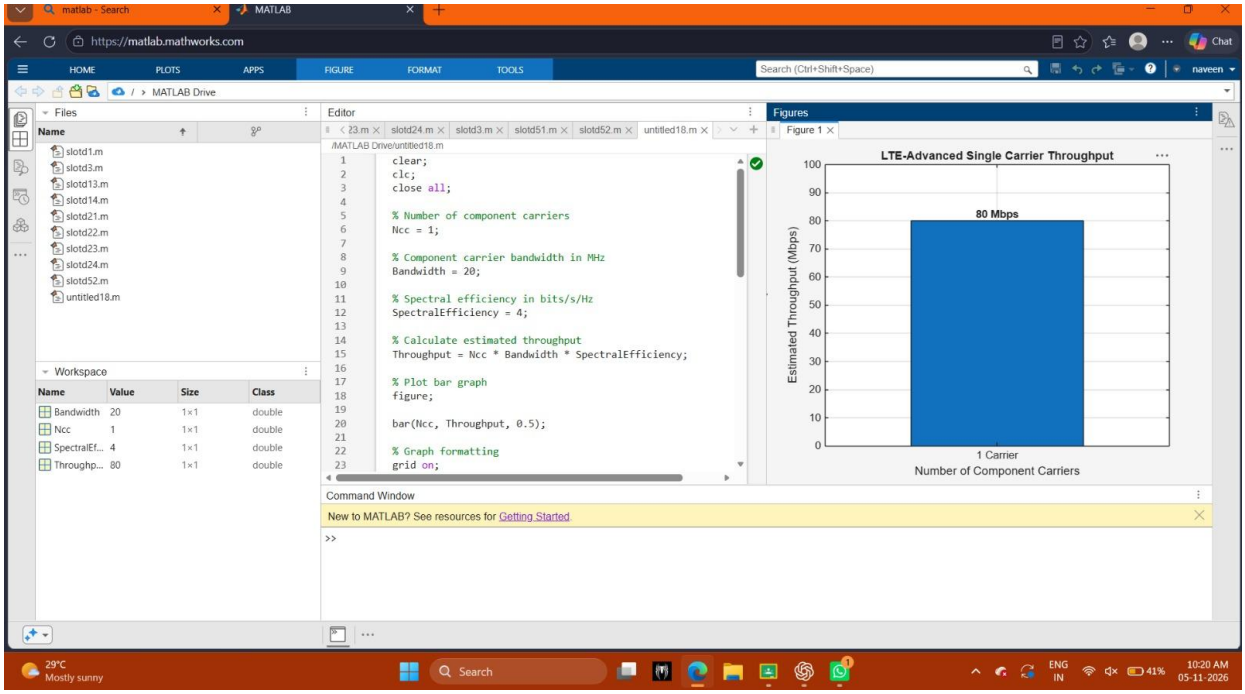
```
clc;
clear;
 close all;
BW1 = 20; % Component carrier bandwidth (MHz)
SE = 4; % Spectral efficiency (bits/s/Hz)
Ncc = 1; % Number of component carriers
BWagg = BW1*Ncc;
Throughput = BWagg*SE; % Mbps for MHz × bits/s/Hz
disp('Single Component Carrier')
disp(['Bandwidth = ',num2str(BWagg),' MHz'])
disp(['Estimated Throughput = ',num2str(Throughput),' Mbps'])
figure
bar(Ncc,Throughput)
grid on
xlabel('Number of Component Carriers')
ylabel('Estimated Throughput (Mbps)')
title('LTE-Advanced Single Carrier Throughput')
```

Expected Output: A graph showing the estimated throughput for one 20 MHz component

carrier. With the selected 4 bits/s/Hz simplified assumption, the estimated throughput is 80

Mbps.

EXPECTED OUTPUT :



OBJECTIVE 2:

PROGRAM CODE :

```
clc; clear; close all;

BWcc = 20; % MHz per component carrier

SE = 4; % bits/s/Hz

CC = [1 2 3]; % Number of component carriers

AggBW = BWcc .* CC; % MHz

Throughput = AggBW .* SE; % Mbps

disp('CC Aggregated BW (MHz) Estimated Throughput (Mbps)')

disp([CC' AggBW' Throughput'])


figure

bar(CC,Throughput)

grid on
```

xlabel('Number of Aggregated Component Carriers')

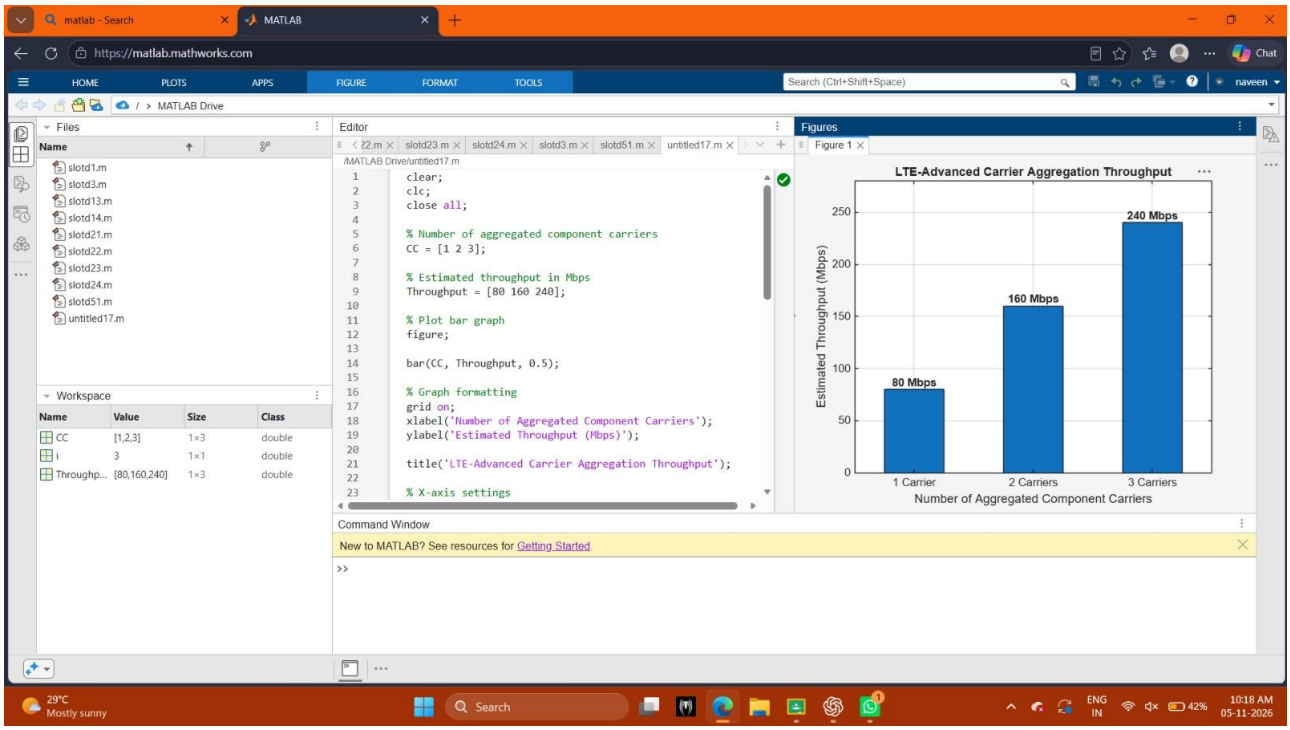
ylabel('Estimated Throughput (Mbps)')

title('LTE-Advanced Carrier Aggregation Throughput')

Expected Output: The graph increases with the number of aggregated component carriers. For

the selected parameters, 1, 2 and 3 carriers provide 80, 160 and 240 Mbps respectively.

EXPECTED OUTPUT :



OBJECTIVE 3:

PROGRAM CODE :

clc; clear; close all;

BW = [5 10 15 20]; % Carrier bandwidths (MHz)

SE = 4; % bits/s/Hz

TP = BW .* SE; % Estimated throughput (Mbps)

disp('Bandwidth (MHz) Throughput (Mbps)')

```
disp([BW' TP'])


figure
plot(BW,TP,'o-','LineWidth',2)
grid on
xlabel('Carrier Bandwidth (MHz)')
ylabel('Estimated Throughput (Mbps)')
title('Throughput vs Component Carrier Bandwidth')
```

Expected Output: Throughput increases as the component-carrier bandwidth increases.

EXPECTED OUTPUT :

